Gauss and Stokes: Surface Geometry, Singular Sources, and Weak Laws

N-20260615

A surface integral can measure mass distributed over a sheet or signed flow through it. In the first case, the metric determines area; in the second, orientation determines the sign. Gauss' theorem and Stokes' theorem then connect these surface measurements to local differential quantities. Their most revealing examples arise when smoothness breaks down: a point charge has zero ordinary divergence everywhere it is defined, yet carries a nonzero flux through every surrounding sphere. The resolution leads directly to weak derivatives and distributions.[1]

§1.1 Area has no preferred orientation

If a surface is parametrized by $r(u, v)$, its scalar area element is

$$dS = \|r_u \times r_v\| \, du \, dv,$$

and the integral of a surface density f is

$$\iint_S f \, dS = \iint_D f(r(u,v)) \|r_u \times r_v\| \, du \, dv.$$

The norm makes this independent of how the parameter coordinates are oriented.

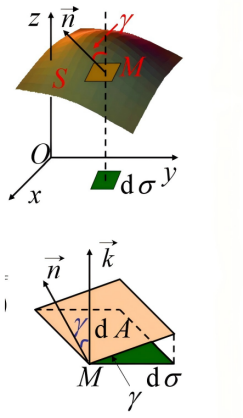


Figure 1.1: A small patch of a surface has area determined by its tangent parallelogram. Its planar projection generally has a different area.

Consider a spherical cap cut from the sphere $x^2 + y^2 + z^2 = R^2$ by $z = h$, with $0 \le h < R$. Parametrize the upper sphere by

$$r(\theta, \varphi) = (R \sin\varphi \cos\theta, \; R \sin\varphi \sin\theta, \; R \cos\varphi).$$

[1] MIT 18.02, supplementary notes V9–V13; Oregon State University, *The Geometry of Static Fields*, §15.2.

The cap corresponds to

$$0 \le \theta \le 2\pi, \qquad 0 \le \varphi \le \arccos(h/R),$$

and $\|r_\theta \times r_\varphi\| = R^2 \sin\varphi$. Therefore

$$\boxed{\text{area(cap)} = 2\pi R^2 \left(1 - \frac{h}{R}\right) = 2\pi R(R-h).}$$

It depends linearly on the cap height $R-h$, although the horizontal cross-sectional radius depends nonlinearly on that height.

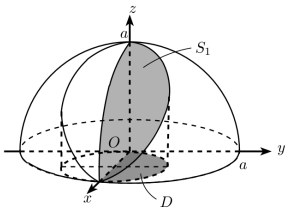


Figure 1.2: Spherical coordinates adapt the parameter domain to the height boundary of a spherical cap.

For a torus with major radius $R > r > 0$, parametrize

$$X(u,v) = ((R + r\cos v)\cos u,\ (R + r\cos v)\sin u,\ r\sin v).$$

Here $u, v \in [0, 2\pi]$, and

$$\|X_u \times X_v\| = r(R + r\cos v).$$

The total area is

$$\boxed{\text{area}(T) = \int_0^{2\pi}\int_0^{2\pi} r(R + r\cos v)\,dv\,du = 4\pi^2 Rr.}$$

The mean of $\cos v$ over a full period is zero, so the long circular direction contributes only its centerline circumference on average.

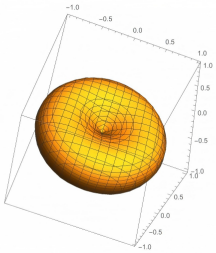


Figure 1.3: The two angular directions of a torus have different local stretch factors, which combine into the area element $r(R + r\cos v)\,du\,dv$.

§1.2 Flux measures outward flow and local sources

For a vector field F, the flux through an oriented surface is

$$\int_S F \cdot n\,dS = \iint_D F(r(u,v)) \cdot (r_u \times r_v)\,du\,dv$$

when $r_u \times r_v$ agrees with the prescribed normal. Unlike scalar area, this changes sign when orientation is reversed.

If $\Omega \subset \mathbb{R}^3$ is bounded by a piecewise smooth closed surface and F is C^1 on a neighborhood of $\overline{\Omega}$, the divergence theorem states

$$\boxed{\int_{\partial\Omega} F \cdot n \, dS = \iiint_{\Omega} \operatorname{div} F \, dV.}$$

The unit normal on the boundary points outward.

For $F(x, y, z) = (x, y, z)$, the sphere of radius R has outward unit normal $n = (x, y, z)/R$, so $F \cdot n = R$. Its total outward flux is

$$R \cdot 4\pi R^2 = 4\pi R^3.$$

The divergence is 3, and integration over the ball gives

$$3 \cdot \frac{4}{3}\pi R^3 = 4\pi R^3,$$

the same answer. The field's local source strength is constant; the boundary records its total over the enclosed volume.

§1.3 Closing Surfaces for Gauss's Theorem

The divergence theorem controls the boundary of a solid. A hemisphere by itself is only part of that boundary. Let S_+ be the upper hemisphere

$$x^2 + y^2 + z^2 = R^2, \qquad z \geq 0,$$

with outward (and therefore upward) normal, and let D be the flat disk in $z = 0$. The boundary of the upper half-ball is $S_+ \cup D$, where the disk carries the *downward* normal $-e_z$. For the field

$$F(x, y, z) = (0, 0, 1 + z), \qquad \operatorname{div} F = 1,$$

the flat cap contributes

$$\iint_D F \cdot (-e_z) \, dS = -\pi R^2.$$

The divergence theorem applied to the half-ball therefore gives

$$\iint_{S_+} F \cdot n \, dS - \pi R^2 = \iiint_{\text{half-ball}} 1 \, dV = \frac{2\pi R^3}{3},$$

and hence

$$\boxed{\iint_{S_+} F \cdot n \, dS = \pi R^2 + \frac{2\pi R^3}{3}.}$$

Omitting the cap would lose the whole term πR^2. As a check, spherical coordinates give $z = R\cos\varphi$, $n_z = \cos\varphi$, and

$$\begin{aligned} \iint_{S_+} F \cdot n \, dS &= 2\pi R^2 \int_0^{\pi/2} (1 + R\cos\varphi) \cos\varphi \sin\varphi \, d\varphi \\ &= \pi R^2 + \frac{2\pi R^3}{3}. \end{aligned}$$

For $F = (x, y, z)$ instead, the cap flux is zero and the hemisphere flux is simply 3 times the half-ball volume, or $2\pi R^3$. The geometry is identical; the field decides whether the missing cap matters.

§1.4 Orientation can turn the same cone flux negative

Consider the lateral surface of the cone $z = \sqrt{x^2 + y^2}$, from its apex to height a, together with the horizontal disk $z = a$. It bounds the solid $\{0 \le r \le z \le a\}$. Write a parametrization of the lateral surface as

$$X(r, \theta) = (r\cos\theta, r\sin\theta, r), \qquad 0 \le r \le a, \quad 0 \le \theta \le 2\pi.$$

The cross product in the parameter order (r, θ) is

$$X_r \times X_\theta = (-r\cos\theta, -r\sin\theta, r).$$

Its vertical component is positive, but its radial component points *into* the cone. Thus it gives the inward normal to the solid, not the outward one. For $F = e_z = (0, 0, 1)$,

$$\iint_{\text{side, up}} F \cdot n \, dS = \int_0^{2\pi} \int_0^a r \, dr \, d\theta = \pi a^2.$$

With outward orientation the side flux is $-\pi a^2$. The top disk, outward oriented upward, contributes $+\pi a^2$. The total outward flux is zero, agreeing with $\operatorname{div} e_z = 0$.

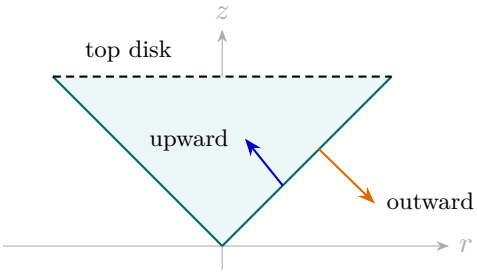


Figure 1.4: In this cross-section of the solid cone, the upward normal on the sloping side points into the solid. The outward normal points downward.

This sign check is also a warning about projecting a curved surface onto a plane. For the constant field e_z, the outward flux through the upper hemisphere is $+\pi R^2$ while through the lower hemisphere it is $-\pi R^2$. Both sheets project to the *same* disk, but their induced orientations are opposite. The two contributions cancel on the complete sphere. The formula $F_z \, dx \, dy$ is an oriented two-form integral; it cannot be used as if $dx \, dy$ were an unsigned surface area element.

§1.5 An inverse-square field exposes a hidden point source

Now let

$$F(x) = \frac{x}{\|x\|^3} \qquad (x \in \mathbb{R}^3 \setminus \{0\}).$$

At every $x \neq 0$, a direct calculation gives

$$\operatorname{div} F = 0.$$

Nevertheless, on the sphere S_R centered at the origin,

$$F \cdot n = \frac{1}{R^2}, \qquad \boxed{\int_{S_R} F \cdot n \, dS = 4\pi.}$$

There is no contradiction with the divergence theorem: the field is undefined at the origin, so it fails the regularity hypothesis on the enclosed ball.

For a region with a small ball around the origin removed, the divergence theorem does apply. Its inner boundary has the normal pointing toward the origin. The outer flux is balanced by an inner flux of -4π. Shrinking the omitted ball retains a nonzero boundary contribution.

This is the mathematical structure of an electrostatic point charge. For charge q and permittivity ε_0, the Coulomb field is $E = \frac{q}{4\pi\varepsilon_0}F$, and every surrounding surface carries flux q/ε_0.

§1.6 Distributional Point Sources

The vector field $F = x/\|x\|^3$ is locally integrable in $\mathbb{R}^3$, so its divergence can be defined as a distribution. For a smooth compactly supported test function φ, put

$$\langle \operatorname{div} F, \varphi \rangle = -\int_{\mathbb{R}^3} F \cdot \nabla\varphi \, dV.$$

Remove a ball of radius ε. Integration by parts on its complement, where $\operatorname{div} F = 0$, gives

$$\begin{aligned} -\int_{\|x\|>\varepsilon} F \cdot \nabla\varphi \, dV &= \int_{\|x\|=\varepsilon} \varphi(x)\frac{1}{\varepsilon^2} \, dS \\ &= \int_{S^2} \varphi(\varepsilon\omega) \, d\omega. \end{aligned}$$

The right-hand side tends to $4\pi\varphi(0)$. Consequently

$$\boxed{\operatorname{div} \frac{x}{\|x\|^3} = 4\pi\delta_0} \qquad \text{in the sense of distributions.}$$

The Dirac delta captures a source concentrated at a point, while the ordinary derivative describes the smooth behavior elsewhere. This identity is the rigorous form of the flux computation.

§1.7 Stokes turns local rotation into boundary circulation

For an oriented surface S with piecewise smooth boundary, Stokes' theorem is

$$\boxed{\oint_{\partial S} F \cdot d\mathbf{r} = \iint_S (\nabla \times F) \cdot n \, dS,}$$

assuming F is C^1 near the surface. The boundary orientation is induced by the surface normal.

A triangle in space is a useful example because its curl flux can be computed without any complicated surface parametrization. Let its ordered vertices be

$$A = (0,0,0), \quad B = (1,0,0), \quad C = (0,1,1)$$

and take $F = (-y/2, x/2, 0)$. Its curl is constant:

$$\nabla \times F = (0,0,1).$$

The oriented area vector of the triangle is

$$\mathbf{A}_S = \frac{1}{2}(B-A)\times(C-A) = \left(0, -\frac{1}{2}, \frac{1}{2}\right).$$

Stokes gives

$$\oint_{A\to B\to C\to A} F\cdot d\mathbf{r} = (\nabla\times F)\cdot \mathbf{A}_S = \frac{1}{2}.$$

One can check this directly. Along AB and CA, the work vanishes because the relevant coordinate is zero. Along BC, write $(x,y,z) = (1-t,t,t)$ for $0\le t\le 1$. Then

$$F\cdot d\mathbf{r} = \left(-\frac{t}{2}, \frac{1-t}{2}, 0\right)\cdot(-1,1,1)\,dt = \frac{1}{2}\,dt.$$

The direct line integral is $1/2$, confirming the sign and area-vector calculation.

The surface spanning the loop need not be flat. Any other smoothly oriented surface sharing the same boundary has the same curl flux, provided the field is smooth on a neighborhood of that surface. Stokes' theorem makes the flux of a curl depend only on the boundary curve.

§1.8 Circulation across Two Surfaces

The line integral in Stokes' theorem may look hard even when the curl is simple. Consider the counterclockwise ellipse

$$C: \quad x = 2\cos t, \quad y = \sin t, \quad z = 0, \qquad 0\le t\le 2\pi,$$

and the field $F = (-y/2, x/2, 0)$. Since

$$\nabla\times F = (0,0,1),$$

the flat elliptical disk $D = \{x^2/4 + y^2 \le 1,\ z = 0\}$, oriented upward, gives

$$\oint_C F\cdot d\mathbf{r} = \iint_D (\nabla\times F)\cdot e_z\,dA = \mathrm{Area}(D) = 2\pi.$$

One can verify the result directly without Stokes. Along C,

$$F = \left(-\tfrac{1}{2}\sin t, \cos t, 0\right), \qquad \mathbf{r}' = (-2\sin t, \cos t, 0),$$

so

$$\boxed{\oint_C F\cdot d\mathbf{r} = \int_0^{2\pi} (\sin^2 t + \cos^2 t)\,dt = 2\pi.}$$

There is even a curved surface with the same boundary: the cap

$$S: \quad z = 1 - \frac{x^2}{4} - y^2, \qquad \frac{x^2}{4} + y^2 \le 1.$$

Its upward vector area element is

$$n\,dS = (-z_x, -z_y, 1)\,dx\,dy = \left(\frac{x}{2}, 2y, 1\right)\,dx\,dy.$$

The vertical curl $(0,0,1)$ therefore sees the same projected area:

$$\iint_S (\nabla\times F)\cdot n\,dS = \iint_D 1\,dx\,dy = 2\pi.$$

These computations make surface independence precise. For this curl field, swapping one spanning surface for another does not change the value; for a *general flux field* it need not be true. Indeed, Gauss says that for two surfaces enclosing a region the flux difference is the integral of $\operatorname{div} F$ over that region. Stokes applies to a curl, whose divergence vanishes identically when the field is smooth. The two theorems thus explain each other's surface-independence hypothesis.

§1.9 The single boundary law behind both theorems

In Euclidean coordinates, associate to $F = (P, Q, R)$ the work one-form

$$\alpha_F = P\,dx + Q\,dy + R\,dz$$

and the flux two-form

$$\beta_F = P\,dy \wedge dz + Q\,dz \wedge dx + R\,dx \wedge dy.$$

Exterior differentiation gives

$$d\alpha_F = \beta_{\nabla \times F}, \qquad d\beta_F = (\nabla \cdot F)\,dx \wedge dy \wedge dz.$$

Thus Stokes' circulation theorem applies the boundary formula to a one-form on a surface, and Gauss' theorem applies it to a two-form on a three-dimensional region:

$$\boxed{\int_{\partial M} \omega = \int_M d\omega.}$$

The difference is the dimension of the manifold M and the degree of the form ω. The orientation of ∂M is induced by that of M.

The algebraic identity $d^2 = 0$ contains the familiar differential identity $\nabla \cdot (\nabla \times F) = 0$. In the smooth setting, it says that an exact flux two-form has zero exterior derivative. Distributional formulations preserve this principle and make it meaningful even when fields are too singular for classical differentiation.

§1.10 Conservation Laws from Divergence

Suppose a time-dependent density $\rho(x, t)$ and flux $J(x, t)$ satisfy

$$\partial_t \rho + \nabla \cdot J = 0.$$

Integrating over a fixed region Ω and applying the divergence theorem gives

$$\boxed{\frac{d}{dt} \int_\Omega \rho(x, t)\,dV = -\int_{\partial\Omega} J(x, t) \cdot n\,dS.}$$

This states that the amount leaving through the boundary accounts for the decrease inside. The same equation describes conservation of mass, charge and probability, with different choices of density and flux.

For a smooth compactly supported test function φ in space, the divergence term can be transferred to the test function:

$$\int_{\mathbb{R}^3} (\nabla \cdot J)\varphi\,dV = -\int_{\mathbb{R}^3} J \cdot \nabla\varphi\,dV.$$

This integration-by-parts identity is the starting point for defining weak solutions when pointwise derivatives do not exist. It is also exactly the definition used above to discover the point source hidden in the inverse-square field.

Smooth surface integration, Stokes' theorem, concentrated charges and weak conservation laws therefore share one mechanism: derivatives describe local changes, while their integrated effect is measured against boundaries. Singularities require a more flexible meaning of derivative, but the boundary accounting remains intact.